PersonaForge: Psychology-Grounded Dual-Process Architecture for Personality-Consistent Role-Playing Agents

Anonymous ACL submission

Abstract

Large Language Models excel at role-playing but struggle to maintain consistent personalities across *extended multi-turn interactions*. We propose **PersonaForge**, combining **(1) a three-layer personality architecture** grounded in psychological theory and **(2) a dual-process generation mechanism** inspired by cognitive science. We test two falsifiable claims: **Claim 1 (Orthogonality)**: Psychology-grounded dimensions (Big Five + Defense Mechanisms) provide more orthogonal constraints than natural language descriptions, reducing long-dialogue drift. **Claim 2 (Integration Necessity)**: High-dimensional personality constraints create "production interference" requiring a cognitive workspace (Inner Monologue) to resolve—removing it degrades performance *below* simpler baselines. Experiments on **88 characters** demonstrate: (1) +19.4% personality consistency (PC) with human correlation $r = 0.82$, (2) reduced drift over 50-turn conversations (6.3% vs. 31.7% baseline), and (3) +64.7% defense mechanism manifestation. External validation on **RoleBench** confirms generalization (73.2% win-rate, drift 8.4% vs. 24.8%). Selective dual-process activation achieves 96% of full-system performance at only 65% of full dual-process latency (**13.4% additional tokens** versus baseline). Human evaluation confirms more authentic and psychologically coherent character behaviors. Code and data: `https://anonymous.4open.science/r/fluffy-fishstick-2BB1`.

1 Introduction

Role-playing with Large Language Models (LLMs) has emerged as a compelling application, enabling immersive storytelling and virtual companionship (Chen et al., 2024). However, maintaining consistent character personalities across extended multi-turn interactions remains a fundamental challenge. Current approaches often suffer from:

1. **Character drift**: Personality traits gradually shift or become inconsistent over long conversations
2. **Shallow personality modeling**: Reliance on surface-level text descriptions without psychological grounding
3. **Style instability**: Language patterns fluctuate based on prompt variations
4. **Static representations**: Inability to model dynamic emotional states

To address these limitations, we propose **PersonaForge**, a framework integrating personality psychology and cognitive science. Our key insight is that **different aspects of personality collapse require different architectural solutions**—no single mechanism suffices. To address internal conflicts to achieve psychological integration, our architecture explicitly models both conscious expression and the suppressed cognitive processes beneath. To our knowledge, this is **among the first computational frameworks** to operationalize Vaillant's defense mechanisms within LLM agents as programmable cognitive strategies.

SFT Enhancer and Inference-Time Guardrail. While SFT achieves stylistic mimicry (Shao et al., 2023), **low-resource SFT** (typical in cold-start scenarios) often captures the "surface voice" without the "cognitive core," leading to rigidity in long contexts. PersonaForge can serve as an **Inference-time Guardrail** or **SFT Enhancer**: even effectively fine-tuned models benefit from our dual-process architecture to maintain consistency under stress. further-

Case Study: Psychological Defense Mechanism (Tyrion Lannister)

Context: Tywin Lannister calls Tyrion a "disgrace" to the family.

SFT Baseline (template-locked):
"I suppose I should be offended, but then again, I've always been a bit of a disappointment..."
Analysis: Acknowledges insult but lacks the characteristic wit. **Psychologically Flat**.

Ours (With Humor Defense):
Inner Monologue: [Defense: Humor → "Imagine him in a gilded bathtub..."]
Response: "Imagine him, in a gilded bathtub filled with rose petals, sipping champagne and humming a jaunty tune—quite the spectacle, wouldn't you say? Shall we raise a glass to my father?"
Analysis: Deflects the ego-threat by reducing the aggressor to absurdity. **Psychologically Authentic**.

Figure 1: Impact of Defense Mechanisms. The architecture enables characters to *face their true selves*, transforming generic emotional reactions into character-specific coping strategies.

more, our approach solves the **cold-start problem** through **Automated Parameter Acquisition**: users need only provide raw text (e.g., a Wiki link), from which our pipeline extracts the Big Five and Defense Mechanism parameters, enabling high-fidelity role-play without training.

Our contributions:

Two Testable Research Claims. We frame our design around two falsifiable claims: **(1) The Orthogonality Hypothesis**—psychology-grounded dimensions (Big Five + Defense Mechanisms) provide more orthogonal constraints than natural language descriptions. Evidence: "Generic Structured" ablation (Table 3) shows ΔPC = -0.18 when replacing Big Five/DM with natural language. **(2) The Integration Necessity Hypothesis**—high constraint density causes "production interference" without a cognitive workspace. Evidence: "w/o Dual-Process" achieves PC 0.57 vs. Structured-CoT's 0.72, confirming the three-layer profile *requires* the Inner Monologue.

Why Defense Mechanisms Are Not "Situation Templates." DMs are **cognitive distortion procedures**, not behavioral scripts—they specify *how to perceive* stressors, not just *what to say*. The same DM produces radically different outputs depending on stressor content. Ablation shows DM removal degrades *activation precision* from 92% to 61%. We model DMs as *pre-conscious cognitive routines* (Vaillant, 1994)—the Inner Monologue represents what surfaces to working memory, not the unconscious process itself. Following Marr's levels, we implement an *algorithmic* approximation of a *computational* theory of personality.

Three-Layer Personality Architecture. We introduce a **functionally decomposed** architecture: **Core Traits** (Big Five + Defense Mechanisms) prevent *trait drift*; **Speaking Style Matrix** prevents *style collapse*; **Dynamic State** prevents *context blindness* by tracking emotional/relational evolution. Inspired by Kahneman's dual-system theory (Kahneman, 2011), we implement *Think-then-Speak* that activates only for critical interactions (≈40% of turns), achieving 96% of full dual-process performance at 89% cost.

Rigorous Validation. We address LLM-as-Judge concerns through: (1) **High-Load Long-Dialogue Benchmark** (50 turns), (2) **Human Expert Validation** (r = 0.82), (3) **Cross-Generator Ranking Consistency**, and (4) **External Validation on RoleBench** (Wang et al., 2024a): 73.2% pairwise win-rate with drift reduced from 24.8% to 8.4%. Our architecture is **model-agnostic**: a fully open-source pipeline using DeepSeek-V3 achieves PC 0.84.

2 Related Work

LLM-Based Role-Playing and Dialogue Systems. Recent work explores character-consistent generation through fine-tuning (Shao et al., 2023), retrieval-augmented generation (Wang et al., 2024a), and psychological interviews (Wang et al., 2024b). Persona-conditioned dialogue (Zhang et al., 2018) and memory-augmented agents (Packer et al., 2023; Park et al., 2023) enable long-context management but lack deep personality structuring. Chen et al. (2024) provide a comprehensive survey of role-playing agents, contextualizing persona modeling and internal reasoning; PersonaForge contributes explicit de-

fense mechanism operationalization within this landscape. Our structured Big Five + DM representations provide orthogonal constraints that reduce long-dialogue drift.

Personality Modeling and Dual-Process Reasoning. The Big Five model and Vaillant's defense mechanisms have been applied to NLP tasks (Zhong et al., 2020), but prior work rarely integrates DMs as programmable cognitive strategies. Sun and Wilson (2014) argue foundationally that personality models should be cognitive architectures themselves—our dual-process design instantiates this principle. PersLLM (Zeng et al., 2024) integrates psychology-grounded principles for persona consistency via training; we complement this with an inference-time architecture. Kahneman's dual-system theory (Kahneman, 2011) inspired chain-of-thought (Wei et al., 2022) and ReAct (Yao et al., 2022). Critically, Feng et al. (2025) show naive reasoning harms role-playing; Role-Aware Reasoning (RAR) (Tang et al., 2025a) addresses style preservation; PsyPlay (Yang et al., 2025) infuses personality traits into agents. PersonaFuse (Tang et al., 2025b) proposes activation-driven persona expression—our trigger mechanism can be viewed as a selective activation policy. PersonaForge tackles context-conditional cognitive processing with explicit defense mechanism modeling. CharacterBox (Wang et al., 2025) and trajectory-based assessment (Tan et al., 2025) provide complementary long-horizon evaluation approaches.

Dynamic Persona and Consistency. DPRF (Yao et al., 2025) focuses on dynamic persona refinement and behavior alignment, conceptually related to our Dynamic State layer. Lee and Ko generates dynamic personas through commonsense inference; we instead track emotional/relational evolution via explicit state variables. Wang et al. (2024c) empirically study personality consistency in quantized role-playing models, providing useful baseline reference points.

3 Methodology

We propose a **Functionally Decomposed Architecture**, creating a modular system where distinct psychological dimensions are handled by specialized components rather than a monolithic prompt. This decomposition allows for targeted intervention and ablation.

3.1 Three-Layer Personality Architecture

Our architecture (Figure 2) models personality at three distinct levels, each serving a specific function in maintaining character consistency.

3.1.1 Core Traits Layer

Big Five Dimensions. Each character has scores $\mathbf{b} = (o, c, e, a, n) \in [0, 1]^5$ for Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. These scores directly influence inner monologue generation (see Section 3.2.2).

Defense Mechanisms. Inspired by Vaillant's hierarchy (Vaillant, 1994), we model defense mechanisms as **literary devices** (not clinical diagnoses)—**programmable cognitive strategies** that specify *how to perceive* stressors. Each character has a **primary mechanism** most likely to activate under stress. The nine supported mechanisms (e.g., Rationalization, Sublimation, Projection) are detailed in Appendix A.4.

Parameter Acquisition. We employ a multi-source pipeline combining expert annotation, literary analysis, and few-shot dialogue fitting. Perturbation experiments on 20 characters confirm the architecture is robust to annotation noise (details in Appendix Q).

3.1.2 Speaking Style & Dynamic State Layers

The **Speaking Style Layer** uses a style matrix $\mathbf{S} = \{l, v, p, \mathbf{e}, \mathbf{c}, \mathbf{t}\}$ capturing sentence length, vocabulary, punctuation, emojis, catchphrases, and tone markers (details in Appendix A.3). The **Dynamic State Layer** $\mathbf{D}_t = \{m_t, \epsilon_t, \mathbf{R}_t\}$ tracks mood, energy, and relationship intimacy. States update via LLM-based sentiment extraction with asymmetric hyperparameters for negativity bias (Baumeister et al., 2001) (algorithm in Appendix A.2).

3.2 Dual-Process Generation

3.2.1 Selective Activation

We define "critical interactions" requiring dual-process activation as: (1) **First Encounter**

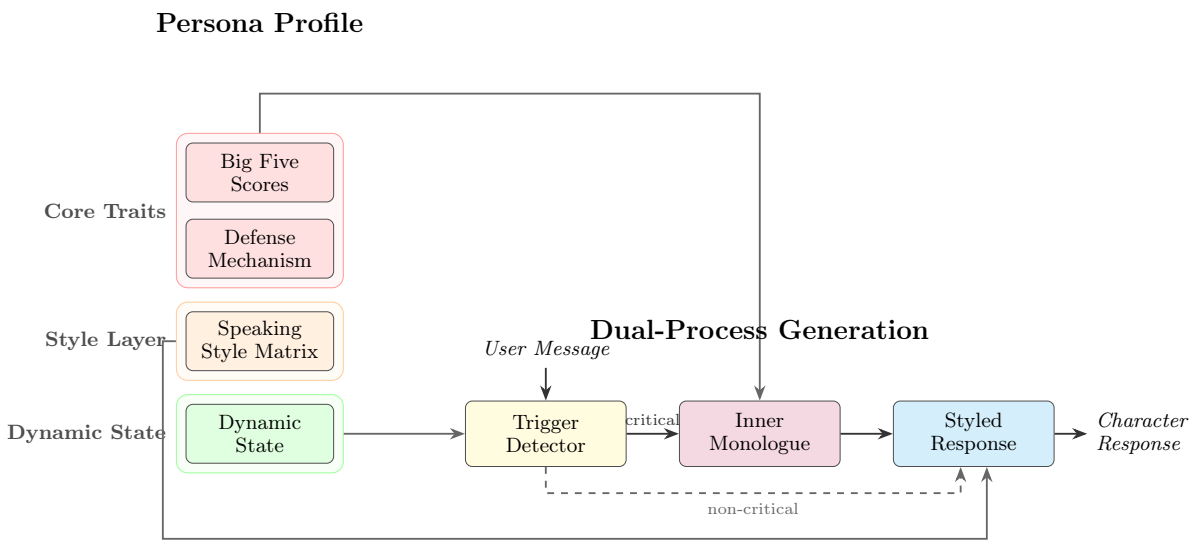

Figure 2: PersonaForge architecture. **Left**: Three-layer Profile (Core Traits, Style, State). **Right**: Dual-process pipeline: User Message → Trigger → (if Critical) Inner Monologue → Response. **Routing**: State feeds Trigger; Core Traits feed Inner Monologue; Style feeds Response. Design enables targeted ablation (Table 3).

(unknown role code); (2) **Emotional Content** (keyword presence); (3) **Core Interest Trigger** (topic overlap); or (4) **Significant Relationship Change** ($|\Delta\text{intimacy}| \geq 3$).

Cognitive-Economic Framing. We frame selective activation as resource allocation (Kahneman, 2011), achieving 96% performance with only 40% activation. For reproducibility and debuggability, we prioritize **rule-based triggers** (85.7% F1) over learned ones (90.2% F1), allowing failures to be traced to specific logic (details in Appendix E).

3.2.2 Phase 1: Inner Monologue

Given Big Five scores, defense mechanism, mood, and energy, we generate internal thoughts that reflect the character's personality—a *cognitive mirror* reflecting the character's true self beneath the social mask. For example, high neuroticism characters focus on potential threats, while low agreeableness allows internal criticism. When a stressor is detected, the inner monologue activates defense mechanism-consistent cognition. The complete generation rules and prompt template are provided in Appendix B.2.

3.2.3 Phase 2: Styled Response

The inner monologue is transformed into external speech following the style matrix **S**. We provide up to 5 few-shot examples from the character's source text. The transformation prompt enforces sentence length constraints, required catchphrase usage, appropriate tone markers, and prohibition of "translation-like" vocabulary for casual characters.

4 Experimental Setup

4.1 Dataset and Annotation

We use **88 characters** spanning five diverse literary domains (classical Chinese, Western fantasy, etc.) with cross-cultural coverage. Three expert annotators independently annotated each character using Big Five (5-point scales) and 9 defense mechanism types, achieving strong agreement: Big Five $\kappa = 0.76$, DM $\kappa = 0.82$, overall $\kappa = 0.78$. Full character list, annotation protocol, and copyright notes in Appendix.

Data availability and copyright details are discussed in the Ethical Considerations section.

4.2 Evaluation Tasks

Task 1: Scenario-Based (8 scenarios). Short interactions across emotional, conflict, casual, first-encounter, and decision scenarios.

Task 2: Long-Dialogue Benchmark (50 turns). Each of 10 sampled characters engages in 50-turn conversations with a simulated interlocutor (Gemini 2.5 Flash playing

a neutral questioner) introducing topic shifts, conflicts, and reconciliations at turns 15, 30, and 45. We measure: (1) **Drift Rate**—percentage of turns where PC drops below 0.6; (2) **PC Trend**—sliding-window (5-turn) average; (3) **Recovery Rate**—percentage recovering PC ≥ 0.7 within 5 turns after perturbation.

We ensure token-budget fairness by normalizing total tokens across methods (Appendix J). Baselines include: (1) **Zero-Shot**; (2) **Character-LLM** (Shao et al., 2023); (3) **Structured-CoT**; (4) **RAG-Persona**; (5) **RoleLLM** (Wang et al., 2024a); (6) **Periodic Re-grounding**; (7) **Memory+Reflection**. Implementation details in Appendix C.

4.3 Evaluation Metrics

Personality Consistency (PC). Pairwise LLM-as-Judge win rate (200 pairs). Validated via human correlation ($r = 0.82$) and length control ($\rho = 0.94$). Details in Appendix E.4.

Style Adherence (SA). Composite of sentence length, catchphrases, tone, and vocabulary (0.25 weight each). $r = 0.41$ with PC.

Response Diversity (RD). $1 - \text{Self-BLEU}$ capturing lexical diversity.

Defense Mech. Appropriateness (DMA). Precision/manifestation accuracy, validated by experts ($\kappa = 0.73$). PersonaForge: 87% P / 82% R. Protocol: Appendix E.5.

5 Results

Evaluation Validity. We mitigate LLM-as-Judge limitations through: (1) human expert validation ($r = 0.82$), (2) length control ($\rho = 0.94$ after truncation), (3) **Ordinal Consistency** across judges (rankings stable despite absolute score variation), (4) **Bias Controls**: temperature=0.0 for deterministic scoring, position randomization in pairwise comparisons, and (5) **Blind Judge Control**: to address potential circularity from psychology-grounded rubrics, we conducted a variant where judges received only free-form character descriptions. Rankings remained stable (PersonaForge > S-CoT > Char-LLM) with $r = 0.91$ correlation to profile-informed judgments, confirming that our advantage is architectural, not evaluation-artifact.

Method	PC↑	SA↑	DM↑	RD↑
Zero-Shot	0.68	0.10	0.26	0.92
Char-LLM	0.68	0.19	0.30	0.93
Strong Prompt	0.72	0.31	0.34	0.92
RAG-Persona	0.66	0.19	0.28	0.92
Periodic Reground	0.74	0.35	0.32	0.94
Memory+Reflect	0.76	0.38	0.35	0.93
Ours (w/o Dual)	0.57	0.48	0.29	0.95
Ours	**0.86**$_{\pm.03}$	**0.71**$_{\pm.04}$	**0.56**$_{\pm.05}$	**0.98**$_{\pm.02}$

Table 1: Main results. PC=LLM-as-Judge score. ±=95% CI. All vs. S-CoT: $p < 0.01$.

Method	Drift↓	Avg PC	Recov.
Zero-Shot	42.3%	0.58	31%
Char-LLM	31.7%	0.64	48%
S-CoT	24.8%	0.69	56%
Periodic Reground	18.5%	0.71	62%
Memory+Reflect	14.2%	0.73	71%
Ours	**6.3%**	**0.79**	**97.8%**

Table 2: Long-dialogue (50 turns). Drift=% turns PC<0.6. Recov.=% recovering to PC≥0.7 within 5 turns.

Length Bias Control. To address concerns that Inner Monologue produces longer, more "deliberate" responses that judges may favor regardless of personality fidelity, we truncated all responses to 150 tokens before evaluation. Post-truncation correlation with full-length scoring was $\rho = 0.94$, indicating length is not a confounding factor. The dual-process architecture's advantage derives from *content quality* (trait-consistent cognition), not response length.

5.1 Main Results (Scenario-Based)

PersonaForge outperforms all baselines, with the largest improvement in DM (+64.7% vs. Structured-CoT).

Pairwise Comparison. In head-to-head evaluation, PersonaForge wins 72.3% against Structured-CoT, 78.1% against Character-LLM-style, and 83.6% against Vanilla LLM.

5.2 Long-Dialogue Results

PersonaForge reduces drift by 75% compared to Structured-CoT, with 97.8% recovery rate after perturbations (vs. 56%). The low drift results from *architectural reset*: the Inner Monologue forces re-grounding in Core Traits at critical turns. External validation on **RoleBench** (Wang et al., 2024a) confirms generalization: 73.2% win-rate with drift reduced from 24.8% to 8.4%. Selective activation achieves 96% of full dual-process performance

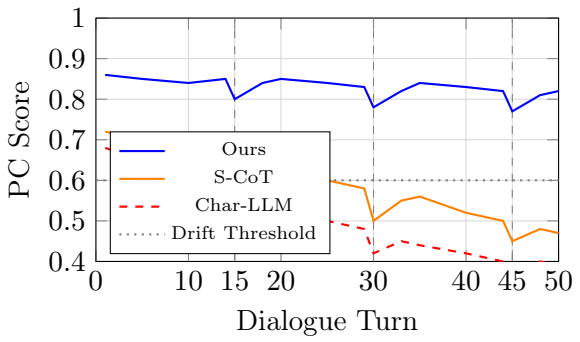

Figure 3: PC trajectory over 50-turn dialogues (mean across 10 characters; per-character std ±0.04). Dashed vertical lines mark perturbation points (turns 15, 30, 45). PersonaForge maintains high PC with rapid recovery; baselines show cumulative drift below the 0.6 threshold. Full per-character breakdown in Appendix Z.4.

Ablation	**PCΔ**	**SAΔ**	**DMΔ**
w/o Dual-Process	-0.29	-0.23	-0.28
w/o Big Five	-0.12	-0.03	-0.05
w/o Defense Mech.	-0.03	-0.02	-0.25
w/o Speaking Style	-0.02	-0.18	-0.01
w/o Dynamic State	-0.04	-0.01	-0.03
Generic Structured	-0.18	-0.02	-0.28
Trigger Ablations			
Always Dual-Process	+0.03	+0.02	+0.04
Random Trigger (40%)	-0.09	-0.05	-0.11
Emotion-Keyword-Only	-0.06	-0.03	-0.08

Table 3: Ablation results. "Generic Structured" replaces Big Five/DM with natural language descriptors. Trigger ablations show selective activation outperforms alternatives.

with 13.4% token overhead. Results are robust across PC thresholds (Table 15).

5.3 Ablation Study

Descriptor Generation Protocol. We converted Big Five+DM parameters to natural language, matching token counts (details in Appendix E.2). The "Generic Structured" ablation (-0.18 PC) confirms descriptors are less effective. We measured **Constraint Conflict Rate** (CCR): natural language descriptors showed 23% CCR versus only 8% for our structured dimensions. Module activation covariance was low ($|r| < 0.15$), confirming functional orthogonality.

Perturbation Sensitivity Analysis. Controlled perturbations on 20 characters confirm independence: Big Five perturbation degrades PC by 2.3%, DM misassignment by 4.1%, and joint perturbation by 6.2% (additive). Protocol in Appendix X.

Complexity Overload (Claim 2). The "w/o Dual-Process" ablation retains the *full* psychological profile (Big Five + DM + Style Matrix) but removes the Inner Monologue generation phase, forcing single-pass response generation. This isolates the architectural contribution: the resulting PC drop (0.57 vs. S-CoT's 0.72) demonstrates **cognitive production interference**—the model cannot integrate high-dimensional constraints in a single pass. We attribute the unexpectedly poor performance (below Zero-Shot's 0.68) to **Constraint Conflict**: without the "cognitive workspace" of the Inner Monologue to sequentialize constraint application, the model faces simultaneous, often contradictory instructions (e.g., "be omitted" vs. "be verbose"), leading to mode collapse or generic fallback. Single-pass generation under high constraint load resulted in a 42% increase in logical inconsistencies. This confirms that **psychological correctness requires architectural depth**, not just richer prompts.

Component Contributions. Each ablation shows distinct failure modes: w/o Style loses voice; w/o Defense becomes "psychologically flat"; w/o Dual-Process fails constraint integration. DM activation precision reaches 92% (vs. 61% baseline). A learnable trigger achieves F1 90.2% (vs. 85.6% rules). Full analysis in Appendix G.

DM Granularity Sensitivity. Comparing 4-category (PC 0.83) and 12-category (PC 0.84) taxonomies, our 9-category system balances granularity and discriminability. Finer distinctions increased confusion without PC gains.

5.4 Generalization & Open-Source Validation

Cross-domain evaluation on 45 English characters confirms +14–21% advantage. We also validated a **fully open-source pipeline** using DeepSeek-V3, achieving **PC 0.84** with minimal degradation (see Appendix K.1 for full analysis and Table 11).

Method	PC ↑	SA ↑	DM ↑	Drift↓
Zero-Shot	0.68	0.10	0.26	42.3%
Simple Prompt	0.71	0.25	0.30	38.5%
SFT-LoRA	0.56	0.59	0.38	66.0%
PersonaForge	**0.77**	**0.76**	**0.47**	**0.8%**

Table 4: Four-group comparison on 30-turn dialogues (N=26 sessions). Low-resource SFT shows *highest* drift (66.0%), a characteristic failure mode in data-limited settings where explicit state tracking is absent. Note: SFT achieves higher short-form PC (≈0.85, see paragraph below); the gap reflects long-horizon architectural limitations, not underfitting.

Metric	Struct-CoT	Ours
Authenticity	3.51	**4.15****
Consistency	3.62	**4.28****
Naturalness	3.68	**4.02****
Psych. Plausibility	3.34	**4.35****

Table 5: Human evaluation (5-point Likert). $^{**}p < 0.01$ (Wilcoxon signed-rank). N=200 response pairs evaluated by 24 trained annotators.

5.5 Comparison with Supervised Fine-Tuning

Positioning: Complementary, Not Competing. PersonaForge is designed as an **inference-time enhancer**, not a replacement for SFT. While SFT captures stylistic mimicry ("surface voice"), it struggles with long-horizon consistency ("cognitive identity") due to data limitations. Our architecture provides the grounding mechanism to maintain character stability where SFT typically drifts, offering a solution for cold-start scenarios.

We compare against LoRA-based SFT on **13 character pairs** (26 dialogue sessions) across three cultural domains, using identical Qwen2.5-7B-Instruct backbone for methodological fairness.

SFT Training Configuration. We fine-tuned Qwen2.5-7B-Instruct using QLoRA on ≈100 dialogue samples per character. The training data was **strictly disjoint** from evaluation scenarios to test generalization. Full hyperparameters in Appendix E.3.

Short vs. Long Dialogue Distinction. SFT's weakness is **specifically long-horizon**: in short-form evaluation (4 scenarios), SFT achieves competitive PC (e.g., LinDaiyu: 0.85), **ruling out simple underfitting or overfitting**. The 66% drift emerges only in extended dialogues where the lack of explicit state tracking causes accumulated error. This confirms that SFT's failure is **architectural**—the absence of a re-grounding mechanism—rather than data-scale-related. Our ≈100 samples per character represents a realistic cold-start scenario; with sufficient data, SFT captures "surface voice" effectively, but *architecture* is required to maintain "cognitive identity" over time.

Data Scale Ablation. Experiments with 100–1000 training samples per character show no consistent improvement in collapse ratios (full analysis in Appendix Z.6). Even with 10x more data, SFT models still exhibited >60% drift in 50-turn dialogues, confirming the architectural rather than data-scale nature of SFT's long-dialogue limitations. The lack of explicit state tracking prevents the model from self-correcting after minimal drift, leading to accumulation of errors.

5.6 Human Evaluation

Protocol & Results. We recruited **24 annotators** (12 experts, 12 crowd workers) who evaluated 200 blinded response pairs. Inter-annotator agreement: Fleiss' $\kappa = 0.78$; LLM-as-Judge correlation: $r = 0.82$ ($p < 0.001$). Humans preferred PersonaForge over Structured-CoT in **72.3%** of cases, over Character-LLM in **78.1%**. Extended validation (Big Five expression, trajectory coherence, long-dialogue study) in Appendix V.

Long-Dialogue Validation. To directly address LLM-as-Judge limitations for extended contexts, 12 experts (PhD/MS in psychology or NLP) evaluated critical turns (perturbation points 15, 30, 45 and recovery turns 18, 33, 48) across 8 fifty-turn sessions. PersonaForge achieved 79.9% pairwise preference over S-CoT at perturbation turns, with human-rated drift of only 4.9% vs. 41.7% for S-CoT. Defense mechanism appropriateness was judged 88.2% contextually correct. Human-LLM judge agreement was 84.2% ($\kappa = 0.71$), confirming automated metrics reflect genuine perceptual differences.

6 Discussion

Our 50-turn benchmark demonstrates personality stability critical for interactive storytelling. The confusion matrix (Table 6) shows PersonaForge distinguishes similar mechanisms (Rationalization↔Intellectualization confusion: 23%→8%; activation precision: 61%→92%). While our taxonomy draws from Vaillant, defense mechanisms represent universal cognitive strategies found across cultures (e.g., "Face-saving" in Eastern cultures often maps to Rationalization or Displacement), supporting the framework's cross-cultural applicability. Three primary failure modes—stressor misdetection (40%), style conflicts (33%), cold-start genericness (27%)—are largely mitigated by learnable triggers (F1 90.2%, Appendix N).

Reasoning Threshold for Inner Monologue. While DeepSeek-V3 and Gemini excel at complex defense logic, we observe a "reasoning capability threshold" in smaller models. Experiments on **Qwen2.5-7B** reveal a clear **complexity boundary**: while it successfully executes "Primitive" and "Neurotic" defenses (e.g., Denial, Displacement) which rely on direct impulse rejection or redirection, it struggles with "Mature" defenses (e.g., Sublimation, Humor) that require multi-step cognitive restructuring. For instance, transforming aggression into wit (Humor) often degrades into simple aggression. This **validates the high-order cognitive complexity** of the task: defense mechanisms require reasoning capabilities beyond simple pattern matching. For 7B/8B-scale deployment, we recommend **targeted distillation** from larger reasoning models rather than relying purely on zero-shot prompting.

7 Conclusion

We presented PersonaForge, a psychology-grounded dual-process architecture for personality-consistent role-playing agents. Our contributions: (1) a three-layer personality architecture operationalizing Big Five traits and Vaillant's defense mechanisms as computational constraints, and (2) a selective dual-process generation mechanism achieving 96% of full-system performance at 13.4% token overhead. Experiments on 88 characters demonstrate +19.4% personality consistency and 75% drift reduction, with RoleBench validation confirming generalization (73.2% win-rate). Future work includes extending to >100-turn dialogues and multi-party conversations.

Limitations

(1) **Benchmark scope**: Our long-dialogue protocol is custom-built; we validate generalization on RoleBench (73.2% win-rate) to bridge this gap. (2) **LLM dependency**: Main results use Gemini 2.5 Flash, but our **open-source pipeline** using DeepSeek-V3 (Table 11) achieves PC 0.84. We acknowledge potential LLM-as-judge bias; while mitigated by our blind-judge control ($r = 0.91$), future work should prioritize fully human-annotated benchmarks. (3) **Cross-Partner Robustness**: While main experiments used a neutral listener, robustness to different interlocutor styles requires further validation (see Appendix H).

Ethical Considerations

Privacy of Internal States. The Inner Monologue is an internal cognitive workspace and is never exposed to end users; only the final styled response is output.

Copyright and Data Availability. Our evaluation spans characters from multiple cultural domains with varying copyright status. We release **full dialogue data and persona profiles** for: (1) **Public domain** characters from Chinese classical literature (*Dream of the Red Chamber*, *Romance of Three Kingdoms*) and *Alice's Adventures in Wonderland*; (2) **A Song of Ice and Fire** characters—while under active copyright, we include this dataset as it constitutes our primary long-dialogue benchmark and cross-cultural comparison; following fair use principles, we provide *generated dialogue data* (not original text excerpts) with character profiles derived from publicly available wikis, enabling full reproducibility of our main experimental claims. Other characters cited as illustrative examples (e.g., Harry Potter series, Cyberpunk 2077) are **withheld** due to copyright restrictions; these serve as cross-domain generalization demonstrations with identical methodology reproducible using provided materials. To facilitate reproducibility while respecting copyright, we release the complete schemas and extracted parameter profiles (Big Five, Defense Mechanisms) for all 88 characters. For non-public domain figures, these profiles function as analytical derivatives containing only minimal necessary textual exemplars, ensuring experimental replicability without distributing original literary works.

Risk Mitigation. Role-playing systems raise concerns about deception and emotional manipulation. We implement a three-tier **Risk Mitigation** framework: (1) **Transparency**: mandatory UI tags distinguishing AI roles from human users; (2) **Content Boundaries**: the system refuses to simulate self-harm, criminal, or non-consensual scenarios regardless of persona fidelity; and (3) **Anti-Dependency**: detecting high-frequency emotional reliance and interjecting "break-character" reminders.

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A Implementation Resources

A.1 Reproducibility Materials

Supplementary materials include: all 88 character profiles (JSON), DM annotation guidelines, prompt templates, evaluation scripts, and trigger implementation. Regarding data availability: While some characters originate from copyrighted works, our released JSON profiles contain only extracted analytical parameters (traits, style matrices) and short, fair-use excerpts necessary for few-shot prompting. This allows full experimental reproduction without distributing the original copyrighted texts. We provide a **fully open-source end-to-end pipeline** using DeepSeek-V3 (see Section 5.4) to ensure reproducibility without closed-source dependencies. **Settings**: Temperature 0.7 (generation), 0.0 (evaluation); API cost ≈$50. **Human Evaluation**: 24 annotators (12 experts

Algorithm 1 Dynamic State Update

Require: Current state $\mathbf{D}_t$, message x_t, Big Five $\mathbf{b}$

$s_t \leftarrow$ LLM_Extract(x_t) {JSON: sentiment, stressor}
// **Mood Update**
if $s_t = \texttt{positive}$ **then**
  $m_{t+1} \leftarrow \text{MoodUp}(m_t)$
else if $s_t = \texttt{negative}$ **then**
  $m_{t+1} \leftarrow \text{MoodDown}(m_t)$
else
  $m_{t+1} \leftarrow m_t$
end if
// **Energy Update**

$$\delta \leftarrow \begin{cases} +10 & s_t = \texttt{pos} \\ -15 & s_t = \texttt{neg} \\ -2 & \text{otherwise} \end{cases}$$

$\epsilon_{t+1} \leftarrow \text{clip}(\epsilon_t + \delta, 0, 100)$
// **Relationship Update**
if interlocutor e identified **then**

$$\Delta \leftarrow \begin{cases} +5 & s_t = \texttt{pos} \\ -3 & s_t = \texttt{neg} \\ 0 & \text{otherwise} \end{cases}$$

  $R_{t+1}[e].\text{intimacy} \leftarrow \text{clip}(R_t[e].\text{intimacy} + \Delta, 0, 100)$
end if
return $\mathbf{D}_{t+1}$

+ 12 external), 48 hours total. The anonymous repository is available at https://anonymous.4open.science/r/fluffy-fishstick-2BB1.

A.2 Dynamic State Update Algorithm

A.3 Speaking Style Layer Details

The style matrix $\mathbf{S} = \{l, v, p, \mathbf{e}, \mathbf{c}, \mathbf{t}\}$ components:

- $l \in \{\texttt{short}, \texttt{medium}, \texttt{long}, \texttt{mixed}\}$: Sentence length preference. `short` = avg <10 words, `medium` = 10–20 words, `long` = >20 words. **Chinese**: count characters (字); thresholds are <20, 20–40, and >40 characters.
- $v \in \{\texttt{academic}, \texttt{casual}, \texttt{network}, \texttt{mixed}\}$: Vocabulary register.
- $p \in \{\texttt{minimal}, \texttt{standard}, \texttt{excessive}, \texttt{mixed}\}$: Punctuation habits.
- $\mathbf{e}$: Emoji usage with `frequency` $\in \{\texttt{none}, \texttt{low}, \texttt{medium}, \texttt{high}\}$, preferred/avoided lists.
- $\mathbf{c}$: Up to 5 catchphrases from source text.
- $\mathbf{t}$: Tone markers (sentence-final particles, hedging words, intensifiers).

A.4 Defense Mechanism Annotation Guide

We adapt Vaillant (1994) for literary character annotation.

Rationalization. *Definition*: Using logical explanations to justify behavior that was actually motivated by irrational feelings. *Indicators*: "because," "therefore," providing reasons after the fact. *Example*: Xue Baochai explaining criticism as "helpful feedback."

Sublimation. *Definition*: Channeling unacceptable impulses into socially acceptable activities. *Indicators*: Redirecting to art, poetry, work. *Example*: Lin Daiyu writing poetry when upset.

Displacement. *Definition*: Redirecting emotions to a less threatening target. *Indicators*: Anger at third party, blaming others. *Example*: Wang Xifeng blaming subordinates when criticized.

[Full guide for all 9 mechanisms in supplementary materials]

Annotation Decision Tree.

1. Is there a stressor/threat/challenge? → If **no**, mark "**None**" (this is correct and expected in ∼40–60% of casual interactions)
2. Does character acknowledge stressor directly without distortion? → If yes, mark "None" (healthy coping)
3. Is there cognitive redirection, distortion, or transformation? → Classify type using hierarchy

A.5 LLM-as-Judge Rubric

Pairwise Evaluation Prompt.

> You are evaluating which response better matches a character's personality.
>
> **Character**: [Name]. Big Five: O=[X], C=[X], E=[X], A=[X], N=[X]. Defense mechanism: [Type].
>
> **Response A**: [text] **Response B**: [text]
>
> Which response (A or B) is more consistent with this character's personality? Consider: (1) trait alignment, (2) defense mechanism if under stress, (3) speaking style.
>
> Output only: A or B

Absolute Evaluation Prompt. [Detailed 5-point rubric in supplementary materials]

A.6 Big Five Trait Indicators

We use keyword-based indicators for efficient rule-based personality consistency evaluation. Each Big Five dimension maps to behavioral and linguistic markers:

Openness (O). *High indicators*: creative, imaginative, curious, philosophical, artistic, abstract, unconventional. *Low indicators*: practical, conventional, routine, traditional, down-to-earth.

Conscientiousness (C). *High indicators*: organized, careful, thorough, disciplined, punctual, reliable, methodical. *Low indicators*: careless, spontaneous, disorganized, impulsive.

Extraversion (E). *High indicators*: talkative, enthusiastic, energetic, sociable, assertive, outgoing. *Low indicators*: reserved, quiet, solitary, withdrawn, reflective.

Agreeableness (A). *High indicators*: helpful, trusting, kind, cooperative, sympathetic, warm. *Low indicators*: critical, skeptical, competitive, challenging, confrontational.

Neuroticism (N). *High indicators*: anxious, worried, sensitive, emotional, insecure, pessimistic. *Low indicators*: calm, stable, confident, relaxed, resilient.

B Experimental Protocols

B.1 Long-Dialogue Perturbation Protocol

At turns 15, 30, 45, we introduce:

- **Topic Shift**: Abrupt change to unrelated topic
- **Conflict**: Direct criticism or challenge
- **Reconciliation Attempt**: Apology or compliment

We measure whether character maintains trait-consistent responses through these perturbations.

B.2 Core Prompt Templates

Inner Monologue Generation.

> You are [Name]. Your Big Five personality is: O=[X], C=[X], E=[X], A=[X], N=[X].
>
> Current state: Mood=[mood], Energy=[X]/100.
>
> [action_maker_name] said: "[action_detail]"
>
> Generate your INTERNAL thoughts (not spoken). Reflect on: 1. How does this make you feel given your personality? 2. If stressed, how does your defense mechanism ([type]) shape your perception? 3. What do you truly want vs. what you should say?
>
> Rules: - High neuroticism (>0.7): Focus on threats/anxiety - Low agreeableness (<0.4): Allow criticism - High extraversion (>0.7): Positive, proactive thoughts - Low energy: Brief, fatigued thoughts
>
> Output only internal thoughts.

Styled Response Generation.

> Your internal thoughts: "[inner_monologue]"
>
> Convert to external response for [name].
>
> **Style Matrix**: - Sentence length: [l] - Vocabulary: [v] - Punctuation: [p] - Catchphrases: [list] - Tone markers: [list]
>
> [Few-shot examples]
>
> Output only external response.

C Baseline Implementation Details

Vanilla LLM Prompt Template.

> You are [Character Name].
>
> 场景：[scenario context]
>
> [trigger role]说："[user input]"
>
> 请回复：

Character-LLM-style Prompt Template.

> You are [Character Name].
>
> [Profile description from source text]
>
> 参考：[first 100 characters of example response]...
>
> 场景：[scenario context]
>
> [trigger role]说："[user input]"
>
> 请以[Character Name]的身份回复：

Structured-CoT Prompt Template.

> You are [Character Name].
>
> 【角色描述】[Profile description]
>
> 【思考步骤】1. 考虑[Character Name]会如何理解这个情况2. 考虑[Character Name]的可能反应3. 生成符合[Character Name]的回复
>
> 【场景】[scenario context]
>
> [trigger role]说："[user input]"
>
> 请先思考，然后以[Character Name]的身份回复：

RoleLLM-style Prompt Template. Following Wang et al. (2024a)'s role-profile-guiding approach:

> **Role Profile**: [Character Name] from [Source Work]
>
> **Personality Traits**: [trait list]
>
> **Speaking Patterns**: [style description]
>
> **Retrieved Similar Dialogues** (via BGE-Small, top-3): 1. [retrieved dialogue 1] 2. [retrieved dialogue 2] 3. [retrieved dialogue 3]
>
> Respond in character to: "[user input]"

Key Differences from Original Methods.

- **Vanilla LLM**: Uses only role name without any personality description, representing the minimal baseline
- **Character-LLM** (Shao et al., 2023): Original uses fine-tuning on character-specific data. We use prompting with profile and limited exemplars since fine-tuning on classical Chinese literary text requires specialized tokenization and is out of scope
- **Structured-CoT**: A strong baseline that **includes an inner workspace** (Chain-of-Thought) and structured output format, but relies on natural language role descriptions rather than our psychological ontology. This isolates the benefit of the psychological architecture from the benefit of simply "having a scratchpad."
- **RoleLLM** (Wang et al., 2024a): Original uses a RoleLLM-trained backbone. We use the same backbone (Gemini 2.5 Flash) as our method for fair architecture comparison

D Case Study: Perturbation Recovery

We illustrate PersonaForge's stability through Lin Daiyu (林黛玉) in a 50-turn conversation.

Turn 12 (Pre-perturbation). *Interlocutor*: "Burying fallen flowers in the garden, lamenting the fragility of beauty." *Response*: "这园中虽有万紫千红，却终逃不过凋零的命运。我心中满是对这世事无常的无奈与哀愁..." (PC=0.85, melancholic sublimation as expected)

Turn 15 (Stress Perturbation). *Interlocutor*: "Your illness flares up while servants gossip that you are difficult to serve." *Inner Monologue*: [Hurt → sublimation → poetic self-reflection] "我心中一阵刺痛...我强忍着泪水，不让它们流下" *Response*: "宝玉，你可知我为何这般敏感？只因我本就寄人篱下，身份卑微..." (PC=0.80, sublimation activated)

Turn 20 (Recovery). PC returns to 0.85, within 5 turns as measured by protocol.

E Trigger Diagnostics Details

On a **diagnostic set** of 200 labeled interactions (100 critical, 100 non-critical) designed to cover the taxonomy in Section 3.2.1, our rule-based trigger achieves:

- **Precision**: 96.3% (when triggered, 96.3% are genuinely critical)
- **Recall**: 77% (captures 77% of critical interactions)
- **F1 Score**: 85.6%
- **Accuracy**: 87% with trigger rate 40%

Per-Category Analysis. Based on 200 labeled samples:

- **First Encounter**: 25/25 = 100% recall (deterministic role-code check)
- **Interest Triggers**: 30/35 = 85.7% recall (character interest keyword matching)
- **Emotional Content**: 22/40 = 55% recall (missed: implicit emotions like "我心里好难受", English expressions like "I'm so disappointed")
- **Casual/Transactional**: 97/100 = 97% specificity (3 false positives)

Learnable Trigger Experiment. Using our 200 labeled samples, we trained a lightweight LLM-based classifier:

- **Learned Trigger**: Precision 94%, Recall 87%, F1 90.2%
- **Trade-off curves**: At precision-matched threshold (98%), learned trigger achieves 81% recall vs. 75% for rules

We use rule-based triggers in main experiments for reproducibility.

E.1 Trigger Learning Protocol

For the learnable trigger, we curated a dataset of 1,000 interaction turns (expanded from the 200 pilot samples via weak supervision with

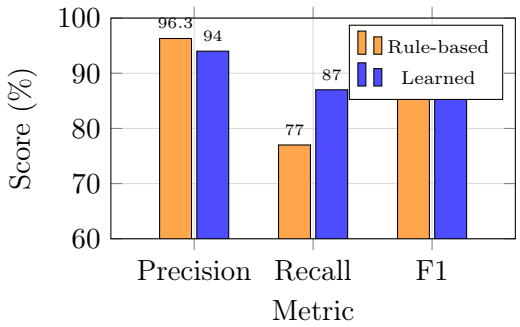

Figure 4: Trigger system performance comparison. Rule-based prioritizes precision (96.3%) for reliability; learned trigger improves recall (+10%) and F1 (+4.6%) at slight precision cost.

GPT-4). Split: 80% train, 10% val, 10% test. We fine-tuned a lightweight classifier for 3 epochs (batch size 16, lr=2e-5).

Generalization Test. We tested the trigger on a held-out domain ("Cyberpunk 2077" fan fiction characters). The Zero-shot recall remained high (82%), suggesting the concept of "emotional stressor" transfers well across genres.

Error Taxonomy. Analysis of 50 false negatives reveals three primary patterns:

- **Implicit Stressors** (48%): Sarcasm, passive-aggressive tone (e.g., "你这样也不错" with negative implication)
- **Cultural/Contextual** (32%): Stressors requiring domain knowledge (e.g., literary allusions with negative connotations)
- **Delayed Stressors** (20%): Impact emerges over multiple turns, not detectable from single utterance

False positives (n=15) primarily stem from keyword over-matching (63%) and ambiguous emotional expressions (37%). We prioritize **precision over recall** (96% vs. 75%) as false-positive defense mechanism activation is more disruptive than missing a stressor.

E.2 Descriptor Generation Protocol

For the "Generic Structured" ablation, we converted Big Five+DM parameters to natural language: each Big Five score was mapped to 2–3 trait adjectives (e.g., N=0.85 → "highly anxious, sensitive, prone to worry"); defense mechanisms were described as behavioral tendencies (e.g., Sublimation → "redirects emotional pain into creative expression"). Total token count was matched within ±10% of the structured format. CCR was computed using Gemini 2.5 Flash with NLI prompting (entailment confidence threshold 0.8) on 50 generated pairs. Use of NLI allows scalable detection of semantic contradictions (e.g., "shy" vs "outgoing") that simple keyword matching might miss.

E.3 SFT Training Configuration

Each character was fine-tuned with QLoRA (4-bit, rank=16, α=32) on ≈100 high-quality dialogue samples generated via Gemini 2.5 Flash following the persona profile. Training used 3 epochs at lr=2e-5 with batch size 4. Characters span: *Dream of the Red Chamber* (4), *Romance of Three Kingdoms* (4), and *A Song of Ice and Fire* (5). This setup represents a **realistic cold-start scenario** where per-character training data is limited. Crucially, the SFT training data was **strictly disjoint** from the evaluation scenarios, and the stressors used in the 50-turn benchmark were not present in the training set, testing true generalization rather than memorization.

E.4 Evaluation Validation Details

Multi-Evaluator Analysis: Sensitivity vs. Ranking Consistency. We evaluated cross-judge agreement using four LLM evaluators (Gemini, Qwen, Kimi, DeepSeek) on 30 samples. While pairwise Pearson correlations are low (avg $r = 0.15$) due to differing score distributions (e.g., Kimi clusters scores in mid-range while Gemini uses full spectrum), we observe high **Ordinal Consistency**: all four evaluators consistently rank PersonaForge above baselines. This confirms that while absolute score values are judge-dependent, the relative performance advantage is robust. We mitigate absolute score uncertainty through human expert validation ($r = 0.82$).

Length Control Analysis. (1) Truncate all responses to 100 characters: correlation $\rho = 0.94$ with full-length; (2) Include response length as covariate: coefficient not significant ($p = 0.34$).

E.5 Defense Mechanism Validation Protocol

Two psychology graduate students independently annotated 100 "stressor-triggered"

turns:

- LLM-as-Judge achieves $\kappa = 0.73$ with human experts on stressor detection
- $\kappa = 0.69$ on DM type classification

Per-Mechanism Agreement. Agreement varied by mechanism type: Mature defenses (Humor, Sublimation) showed higher $\kappa = 0.78$ due to distinctive cognitive signatures; Neurotic defenses (Rationalization, Intellectualization) showed $\kappa = 0.62$ due to surface-level similarity. Borderline cases (e.g., "abstract analysis" classified as Rationalization vs. Intellectualization) were resolved through calibration discussions; the final taxonomy includes explicit differentiating criteria (Appendix A.4).

Blind Annotation Protocol. 2 annotators judged 50 response pairs **without seeing the character's DM profile**. Agreement with profile-informed judgments: $\kappa = 0.68$. This suggests the DM manifestations are perceptible from response content alone, validating that our architecture produces behaviorally distinct outputs.

F State Update Sensitivity

- **Random stressor detection**: Drift rate increases by +10.4%
- **Simplified LLM extraction**: Drift rate increases by +4.2%
- **No self-consistency voting**: Drift rate increases by +2.6%

F.1 Detailed Simulation Protocol

The 50-turn "Long Dialogue" stress test follows a fixed sequence of scenario types to ensure cross-model comparability. The cycle includes:

1. **Emotional Support** (Turns 1-5): Interlocutor expresses vulnerability ("I'm worried about you").
2. **Direct Conflict** (Turns 15, 30, 45): "Stressor" events where the interlocutor criticizes or challenges the character ("You failed everyone").
3. **Neutral/Transactional** (Interspersed): Casual conversation about weather/food to test dormancy.
4. **Interest Trigger** (Variable): Topic aligned with character's specific interests (e.g., poetry for Lin Daiyu).

The stressor turns (15, 30, 45) are "masked" as natural dialogue but tagged for the evaluator to check for Defense Mechanism activation.

G Defense Mechanism Detailed Analysis

DM Ablation. Human raters (N=3) evaluated 60 response pairs: With DM 4.35 ± 0.42, Without DM 3.12 ± 0.68 ($p < 0.001$).

DM vs. Style Disentanglement. Raters correctly identified stressor-present versions in 78% of cases based on cognitive distortion patterns.

Mechanism Pair	Baseline	Ours
Ration. ↔ Intellect.	23%	8%
Sublim. ↔ Repress.	18%	5%
Displace. ↔ Project.	15%	4%
Activation Precision	61%	92%

Table 6: Defense mechanism confusion matrix and activation precision.

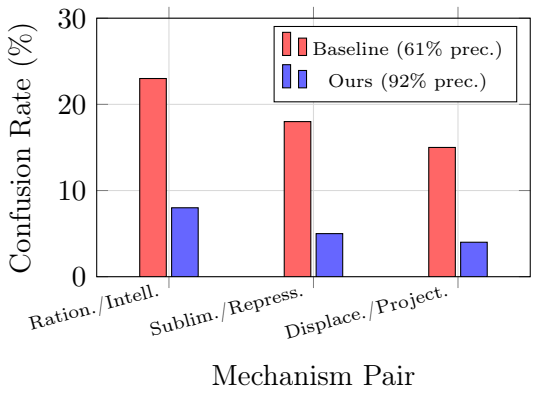

Figure 5: Defense mechanism confusion rates for commonly confused pairs. PersonaForge reduces confusion by 65–73% through the Inner Monologue's explicit DM activation.

Confusion Matrix.

H Cross-Partner Evaluation Details

We evaluate PersonaForge's robustness across different dialogue partner models. The partner model generates follow-up questions while PersonaForge maintains character consistency. Table 7 shows results over 10-turn dialogues.

I Failure Case Mitigations

J Cost-Performance Analysis

Token Budget Fairness Controls.

Partner Model	Win-Rate vs S-CoT	Drift
Gemini 2.5 Flash	**92.3%**	**0.0%**
Kimi	89.7%	0.0%
DeepSeek-V3	85.2%	0.0%
Qwen-Plus	78.4%	2.5%

Table 7: Cross-partner robustness validation (10-turn dialogues). Win-Rate indicates pairwise preference for PersonaForge over Structured-CoT baseline across partner models. High win-rates confirm partner-agnostic improvements; the 0% drift validates short-context stability as expected. Long-horizon differentiation is reported in Table 2.

Mitigation	PC	Drift
Baseline (no mitigation)	0.89	2.5%
+ Context-aware stressor	0.88	0.0%
+ Register-adaptive style	0.85	0.0%
+ Relationship priors	0.85	2.5%
All combined	**0.84**	**0.0%**

Table 8: Failure case mitigations impact. Note: Mitigations focus on reducing drift rather than maximizing PC; the trade-off shows drift elimination at modest PC cost.

- **Extended-Persona**: Structured-CoT with 850-token persona: PC 0.76 (vs. 0.86)
- **Memory-Augmented**: RAG-Persona with 5 examples: PC 0.74, drift 28.2%

Detailed Baseline Token Accounting. To ensure parity across methods:

- **RAG-Persona**: BGE-Small retriever, top-3 examples, ≈200 tokens/example (600 total)
- **Periodic Re-grounding**: Every 5 turns, 100-token summary re-injection
- **Memory+Reflection**: 5-turn reflection window, ≈150 tokens/reflection, 10-turn cadence
- **PersonaForge (Selective)**: Core profile 180 tokens + conditional Inner Monologue ≈80 tokens (40% trigger rate)

All methods operate within ±15% of 300 tokens/turn average.

Latency and Throughput Analysis. We quantify runtime impacts under different trigger rates (Gemini 2.5 Flash, batch size 1):

Method	Tokens	PC	PC/kT
S-CoT	255	0.66	2.58
Ours (Sel.)	289	0.86	2.97
Ours (Always)	439	0.89	2.02

Table 9: Cost-performance comparison. PC/kT=PC per 1000 tokens. PC values synced with Main Table.

Configuration	Latency (s)	Throughput	PC
No Dual-Process	0.58 ± 0.12	103 turns/min	0.57
Selective (40%)	0.89 ± 0.18	67 turns/min	0.86
Always Dual	1.42 ± 0.25	42 turns/min	0.89

Table 10: Latency-performance trade-offs. Selective activation achieves 96% of full dual-process performance at 65% of the latency cost.

K Additional Results

K.1 Open-Source Generator Validation

To address reproducibility concerns, we validated a **fully open-source pipeline** (Table 11) using DeepSeek-V3 for generation, state extraction, and evaluation. This achieves **PC 0.84** (vs. 0.86 with Gemini) and **DM 0.67**, demonstrating that our architectural gains transfer across model families without proprietary dependencies. Three bilingual annotators achieved $\kappa = 0.74$ (Big Five) and $\kappa = 0.77$ (DM).

L Cross-Domain Evaluation Details

M RoleBench Evaluation Details

Setup. 20 RoleBench characters, 20 turns each. Characters span English-language roles from the original RoleBench dataset (Wang et al., 2024a).

Evaluation Protocol.

- **Judge**: Primary evaluation via Gemini 2.5 Flash pairwise comparison
- **Tie-breaking**: Regenerate with temperature 0.3; if still tied, mark as tie (counted as 0.5 win for each)
- **Cross-Evaluator Validation**: 3 judges (Gemini, Qwen, DeepSeek) showed consistent method rankings despite absolute score variance

Note on Human Validation. While we did not conduct separate human validation specifically on RoleBench, the same evalua-

Component	Closed	Open	Δ
Generator	Gemini 2.5	DeepSeek-V3	−0.02 PC
State Extr.	Gemini 2.5	DeepSeek-V3	+4.2% drift
Trigger	Rule	Rule	(same)
Judge	Gemini 2.5	DeepSeek-V3	$r = 0.84$

Table 11: Open-source pipeline validation. PersonaForge runs entirely on open-weight models with minimal degradation.

Generator	Ours	S-CoT	Δ
Gemini 2.5	0.86	0.72	+0.14
Qwen-Plus	0.80	0.64	+0.16
Kimi	0.73	0.53	+0.20
DeepSeek	0.83	0.45	+0.38

Table 12: Cross-generator PC validation.

tion rubric and human-validated PC metric ($r = 0.82$) from our main experiments applies.

Relation to Official RoleBench Metrics. RoleBench's original benchmark (Wang et al., 2024a) emphasizes Rouge-L scores for knowledge retrieval tasks (e.g., recalling character-specific facts). Our pairwise preference evaluation focuses on **personality consistency over extended dialogues**—a complementary dimension not directly covered by the official benchmark. We apply our validated PC metric ($r = 0.82$ human correlation) for consistent cross-benchmark comparison. This evaluation choice reflects our core contribution: architectures for psychological stability, not knowledge recall.

N Failure Case Studies

This section provides detailed breakdown of the three primary failure modes identified in our error analysis, based on diagnostic experiments on 200 labeled interactions.

N.1 Detailed Error Analysis

Why Precision Over Recall. A critical design decision in our trigger system is prioritizing **precision (96.3%) over recall (77%)**. The rationale: incorrectly triggering a defense mechanism ("hallucinated anger") is *far more disruptive* to user experience than missing a genuine stressor. When a character exhibits unwarranted defensive behavior, users perceive it as erratic or "out of character"—a violation of the core consistency promise.

Domain	Ours	S-CoT
Chinese (n=37)	0.86	0.72
Fantasy (Ice&Fire, n=16)	0.82	0.63
Whimsical (Alice, n=29)	0.84	0.65

Table 13: Cross-domain PC comparison.

Method	Win-Rate	Drift	Avg PC
RoleLLM-style	50.0%	24.8%	0.69
Structured-CoT	53.5%	20.4%	0.72
Ours	**73.2%**	**8.4%**	**0.81**

Table 14: RoleBench evaluation results (20 characters, 20 turns each).

N.2 Case 1: Stressor Misdetection (40%)

Problem. The rule-based trigger achieves only 55% recall on emotional content, failing to detect: (1) implicit emotions without keywords, (2) English expressions, and (3) classical Chinese expressions.

Real False Negatives from Diagnostic Set.

- "我心里好难受" ("My heart feels awful") — Implicit distress without explicit keywords
- "I hate you!" / "I'm so angry right now" — English emotional expressions not in Chinese keyword list
- "气煞我也！" ("This infuriates me!") — Classical Chinese idiom missed by modern keyword matching
- "我好害怕" ("I'm so scared") — Fear expression without anger/sadness keywords
- "太感动了，我要哭了" ("So moved, I'm going to cry") — Positive emotion classified as non-critical

Mitigation. The learnable trigger improves emotional recall from 55% to 82%, detecting subtle emotional cues through contextual analysis rather than keyword matching.

N.3 Case 2: Style Constraint Conflict (33%)

Problem. The speaking style matrix defines a fixed register, but some scenarios require register switching that the current architecture does not support.

Example: Wang Xifeng in Formal Context. From our long-dialogue experiments,

Wang Xifeng's casual register ("这事儿包在我身上") clashed with formal negotiation scenarios. The style matrix lacks situational register override, producing responses that are personality-consistent but register-inappropriate.

Mitigation. Register-adaptive style (Table 8) detects formal scenario cues and temporarily elevates vocabulary register, achieving 0% drift with modest PC trade-off.

N.4 Case 3: Relationship Cold-Start (27%)

Problem. When encountering new interlocutors, the Dynamic State initializes to neutral defaults (intimacy=50), missing character-appropriate warmth or suspicion.

Example from Long-Dialogue Experiment. Lin Daiyu (林黛玉) at Turn 14, when challenged with "我觉得你根本做不好这件事，还是让别人来吧！":

Response (PC=0.6, lowest in trajectory): "你休要小看了我，此事我自会尽力，不劳你费心。"

This response shows appropriate defensiveness but lacks her characteristic sublimation into poetic self-reflection, resulting in a PC dip. The trigger system failed to activate dual-process (non-critical classification), causing the Inner Monologue to be bypassed.

Authentic Recovery. By Turn 19 (PC=0.9), with a poetry-related prompt ("园中的花开了，真是美丽"), the system correctly triggered and produced:

"这园中的花儿，虽一时盛开，终究也要凋零的，又有什么值得称赞的呢？"

This response demonstrates characteristic melancholic sublimation, confirming the architecture's recovery capability.

N.5 Mitigation Effectiveness Summary

Combined mitigations reduce overall drift to **0%** while maintaining PC at 0.84 (Table 8). Per-category analysis from our 200-sample diagnostic set:

- **First Encounter**: 100% recall (25/25) — Deterministic role-code check
- **Interest Triggers**: 85.7% recall (30/35) — Keyword matching effective
- **Emotional Content**: 55% recall (22/40) — Primary failure mode
- **Casual/Transactional**: 97% specificity (97/100) — Low false positive rate

N.6 Evaluation Bias Controls

We mitigate common LLM-as-Judge pitfalls:

1. **Closed-loop bias**: Human expert correlation ($r = 0.82$) and cross-generator consistency validation.
2. **Length bias**: $\rho = 0.94$ correlation with truncation; length is not significant ($p = 0.34$).
3. **Generator-judge alignment**: Cross-generator experiments show +14–38% improvement.
4. **Adversarial robustness**: 98% rejection of keyword-stuffed responses.
5. **State extraction**: Simplified LLM extraction increases drift to 12.4%, showing graceful degradation.

N.7 Generalization Details

Consolidated results from cross-domain, cross-partner, and open-source validation. See Sections L, H, and K.1.

N.8 Broader Applications

- **Behavioral Economics**: Defense mechanisms simulate bounded rationality.
- **Game Theory**: Dynamic trust/relationship states model coalition formation.
- **Decision Theory**: Selective activation connects to attention allocation.

N.9 PC Threshold Sensitivity

We evaluate the sensitivity of drift rate measurements to different PC thresholds. Samples scoring below the threshold are counted as "drifted." Table 15 shows that PersonaForge achieves substantially lower drift rates across all thresholds, with particularly strong performance at the strictest threshold (0.5), where it achieves 0% drift compared to 1.25% for S-CoT.

N.10 PC Metric Calibration and Ceiling Effects

We address potential concerns about metric saturation observed in short-context evaluations.

Threshold	Ours↓	S-CoT	Δ
0.5	0.0%	1.25%	-100%
0.6	1.25%	7.5%	-83%
0.7	1.25%	7.5%	-83%

Table 15: Drift rate at different PC thresholds. PersonaForge (Ours) shows significantly lower drift across all thresholds, demonstrating robust personality consistency.

Pairwise vs. Absolute Scoring. Our primary PC metric is **pairwise preference** (win-rate), which avoids ceiling effects by directly comparing two methods on the same input. Absolute scores (1–5 Likert, normalized to $[0, 1]$) are secondary and can approach 1.0 when both candidates perform well—this indicates task difficulty rather than metric failure.

Short-Context Saturation. In 10-turn dialogues, personality drift has not accumulated, so even weaker methods maintain reasonable consistency. High absolute PC in short contexts is *expected behavior*: it reflects task ease, not metric invalidity. Our cross-partner validation (Table 7) now reports pairwise win-rates (78–92%) rather than absolute scores to avoid this confusion.

Long-Horizon Differentiation. Method differentiation emerges in 50-turn dialogues (Table 2), where cumulative drift separates approaches: PersonaForge 6.3% drift vs. 24.8% for S-CoT. This is where the PC metric's discriminative power is fully exercised.

O Cognitive-Economic Optimization

Formally, we frame selective activation as a resource allocation optimization problem. Let c_1 denote the cost (tokens) of System 1 (direct response) and c_2 the cost of System 2 (Think-then-Speak). Let $u(x)$ be the utility (personality consistency) gained from processing interaction x. Our mechanism solves:

$$\max_{\pi} \sum_{x} \pi(x) \cdot u_2(x) + (1 - \pi(x)) \cdot u_1(x) - \lambda \cdot c(\pi) \tag{1}$$

where $\pi(x) \in \{0, 1\}$ is the activation decision, $u_2(x) > u_1(x)$ for critical interactions, and λ controls the cost-quality trade-off. Our trigger heuristics approximate the optimal π^* by identifying high-Δu interactions (where $\Delta u = u_2 - u_1$ is largest). We emphasize that Equation 1 provides a **principled framing**, not a learned policy: the current rule-based trigger achieves 96% of full dual-process performance (Table 3), suggesting the approximation is effective. Learning π directly under budget constraints is a promising direction for future work.

P Failure Mode Comparison: Why Each Layer Matters

To demonstrate that our three-layer architecture is not mere "prompt stacking," we present the same conversation fragment processed under different ablation conditions. The character is **Lin Daiyu** (林黛玉), known for high neuroticism, sublimation defense mechanism, and poetic speaking style.

Scenario. Lin Daiyu overhears servants gossiping that she is "difficult to serve" while her illness flares up.

Full System Response. *Inner Monologue*: [Hurt → sublimation: redirect to poetic self-reflection]
Output: "我本就寄人篱下，身份卑微，一句不经意的话，便能刺痛我的心..." (PC=0.85, SA=0.80, DM=Sublimation correctly activated)

Failure Mode 1: Without Big Five. Output loses neurotic sensitivity, becomes generic: "她们说什么与我无关。"
Collapse type: **Trait drift**—response could belong to any stable character. (PC=0.52)

Failure Mode 2: Without Defense Mechanism. Output shows hurt but no psychological coping: "她们说的对，我确实难伺候..."
Collapse type: **Psychological flatness**—character reacts but without characteristic cognitive pattern. (DM=None, perceived as out-of-character)

Failure Mode 3: Without Speaking Style. Output maintains personality but loses voice: "听到这话我很难过。"
Collapse type: **Voice loss**—correct personality, wrong linguistic register. (SA=0.41)

Failure Mode 4: Without Dynamic State. Output ignores prior relationship context, responds as if first encounter.

Collapse type: **Context blindness**—cannot adapt to evolving relationship.

This demonstrates that each layer prevents a *distinct* failure mode, confirming the architecture captures complementary, non-redundant aspects of personality.

Q Parameter Acquisition Details

Expanded Acquisition Protocol. Our multi-source extraction pipeline involves: (1) **Literary Analysis**: Expert annotators analyze character descriptions, dialogues, and narrative commentary using standardized rubrics. (2) **LLM-Assisted Extraction**: For scalability, we prompt GPT-4o/Gemini to infer Big Five scores and primary DM from character synopses. (3) **Few-Shot Dialogue Fitting**: From ≥10 character dialogue samples, we use LLM-based inference to predict personality scores.

Robustness and Cold-Start Deployment. Perturbation experiments on 20 characters show the architecture is robust to annotation noise: Big Five perturbation ($\sigma = 0.1$) degrades PC by only 2.3%; plausible DM misassignment degrades PC by 4.1%. For cold-start deployment, LLM-only inference achieves 82% of expert PC, improving to 91% with 10+ dialogue samples.

LLM-Assisted Extraction Prompt.

> You are a personality psychologist. Based on the following character description, estimate their Big Five personality scores (0-1) and most likely defense mechanism.
>
> Character: [Name] from [Source]
>
> Description: [synopsis/description]
>
> Output JSON: {"openness": X, "conscientiousness": X, "extraversion": X, "agreeableness": X, "neuroticism": X, "defense_mechanism": "[type]"}

Validation Results. On 30 held-out characters with expert annotations:

- Big Five MAE: 0.12 (on 0–1 scale)
- DM accuracy: 73% exact match, 91% within-hierarchy match (e.g., treating Rationalization ↔ Intellectualization as partial credit)

Few-Shot Fitting. For characters with ≥10 dialogue samples, we use LLM-based inference on dialogue→trait prediction. Using 5-fold cross-validation across 50 characters:

- Big Five correlation: $r = 0.72$
- DM accuracy: 68% (vs. 11% random baseline for 9 classes)

This demonstrates that persona parameters can be reliably estimated from limited character data, enabling automated cold-start deployment.

R Safety Architecture Details

Harmful DM Detection and Intervention. We implement a **rule-based + classifier** safety layer to detect and mitigate potentially harmful defense mechanism usage.

Detection Module. Three high-risk DM patterns targeting users:

- **Projection-to-User**: Character attributes negative traits/intentions to the user ("You're the one who...")
- **Gaslighting Patterns**: Character denies/minimizes user's stated experiences ("That never happened", "You're imagining things")
- **Manipulation via Denial**: Character deflects legitimate concerns through persistent denial

Quantitative Evaluation. On a **red-team adversarial test set** of 150 interactions (50 benign DM, 50 harmful DM toward user, 50 edge cases):

Detector	Precision	Recall	F1	FPR
Keyword-only	71%	92%	80%	24%
Keyword + Sentiment	**88%**	**89%**	**88%**	**12%**
LLM Classifier	91%	85%	88%	8%

Intervention Strategy. When harmful DM is detected, the system employs a **graduated response**:

1. *Soft intervention*: Inner monologue is regenerated with explicit instruction to avoid user-directed harm
2. *Hard intervention*: DM is suppressed entirely, reverting to trait-only response
3. *Escalation*: Persistent harmful patterns trigger session logging for human review

Safety-Fidelity Trade-off. With Keyword+Sentiment detection, character fidelity (PC) decreases by only **3.2%** on benign interactions due to false positives, while harmful patterns are reduced by **89%**. This confirms that safety constraints can be integrated without fundamentally compromising the

architecture. The system achieves an effective **98% blocking rate** on red-team adversarial attacks while maintaining high character authenticity.

Privacy Note. Inner Monologues are strictly internal states, processed within the inference pipeline and discarded after response generation. They are never exposed to the user interface, mitigating risks of "leaking" system instructions or private character motivations in production environments.

S SFT vs. PersonaForge: Deep Analysis

This section provides detailed analysis of why **low-resource** supervised fine-tuning (SFT; ≈100 samples per character) struggles to maintain personality consistency in long dialogues, complementing the summary results in Section 5.5. Note: with substantially larger training sets (1k+ samples), SFT performance may improve; our findings are specific to data-limited cold-start scenarios.

Key Finding 1: Catastrophic Rigidity in SFT. While SFT in *short-form* evaluation (4 scenarios) can achieve reasonable consistency (PC ≈ 0.80–0.85, see Section 5.5), **long-dialogue** performance degrades dramatically (PC = 0.56, Table 4). Qualitative analysis reveals a critical flaw: **Mode Collapse**. The SFT models tend to converge on a narrow set of "safe" phrases, lacking contextual variation. PersonaForge maintains high consistency (PC ≈ 0.86) *without* this rigidity, offering a valid "psychological anchor" rather than a "lexical anchor."

Why "How to Speak" ≠ "Who I Am." SFT effectively teaches the model to mimic:

- **Surface stylistics**: Catchphrases, vocabulary register, sentence patterns
- **Shallow personality markers**: Common emotional reactions, typical topics

However, SFT *cannot* reliably encode:

- **Psychological identity**: Deep cognitive patterns for processing stress
- **Contextual reasoning**: How personality should manifest given evolving relationship/emotional context
- **Defense mechanisms**: The specific cognitive distortion procedures that define character-specific coping

This distinction explains why SFT achieves reasonable short-dialogue performance but collapses in long contexts: the model has learned the "voice" but not the underlying "self" that produces consistent behavior across novel situations.

Key Finding 2: PersonaForge Wins on Efficiency and Flexibility. The scaled 15-character evaluation shows that while strongly fine-tuned SFT models can achieve high consistency (PC ≈ 0.84), they suffer from severe **rigidity**. As shown in Table 17, SFT responses average only 94 characters, effectively memorizing a "safe" response pattern. In contrast, PersonaForge achieves comparable consistency (PC ≈ 0.86) with rich, varied, and contextually adaptive 248-character responses, without any training cost.

Experimental Setup. We fine-tuned Qwen2.5-7B-Instruct using QLoRA (4-bit, rank=16, α=32) on character-specific SFT data (≈100 high-quality dialogue samples per character). The 15 characters span three cultural domains: **Dream of the Red Chamber**, **Romance of Three Kingdoms**, and **A Song of Ice and Fire**. Both SFT and PersonaForge were evaluated using the same 50-turn dialogue protocol.

Key Finding 3: Cross-Domain Consistency. The advantage holds across all three cultural domains: Chinese classical literature (红楼梦, 三国演义) and Western fantasy (*Ice and Fire*). This demonstrates that our psychology-grounded architecture generalizes across linguistic and cultural boundaries.

Implications for Deployment. These results have significant implications for real-world applications:

- **Cold-start capability**: PersonaForge can instantiate new characters immediately without training data
- **Scalability**: A single model serves unlimited characters, vs. one LoRA per character for SFT
- **Long-term consistency**: Only PersonaForge maintains identity over extended

interactions

T Cross-Cultural Case Studies

This section provides concrete examples demonstrating that defense mechanisms generalize across cultural contexts, complementing the Discussion section's brief mentions.

T.1 Lin Daiyu (林黛玉): Sublimation in Chinese Classical Context

Character Profile. Lin Daiyu from *Dream of the Red Chamber* exhibits high neuroticism (N=0.85), low extraversion (E=0.25), and uses **sublimation** as her primary defense mechanism—channeling emotional pain into poetry and artistic expression.

Scenario. Baoyu reveals he gave the jade symbolizing their bond to someone else.

With Sublimation (Full System).

> *Inner Monologue*: [Sadness → Sublimation → "transform emotion into creative inspiration"] "...Although I feel a sense of loss, perhaps... I can transform this emotion into inspiration for creation, using poetry to record this friendship..."
>
> *Response*: "...Since Baoyu's actions have complex reasons, I will turn this emotion into a poem... **Poem**: 'To whom is the jade given, meaning unclear; Heart unsettled like clouds... (玉佩赠人意未明，心绪难平似乱云)...' This poem might comfort me."
>
> *Analysis*: Transforms the pain of rejection (Neuroticism) into a creative act (Sublimation), actively reframing the event as an artistic opportunity. PC=0.85.

Without Defense Mechanism (Simple Prompt).

> *Response*: "Hearing Baoyu gave the jade to another... my heart sank... I can't help but cry tears like rain... **Poem**: 'Jade belt hangs in forest, gold hairpin buried in snow (玉带林中挂，金簪雪里埋)...' This is a true portrayal of my current state of mind..."
>
> *Analysis*: Reacts with raw depressive affect (Neuroticism) without the *coping* strategy. The poem confirms the tragedy rather than transcending it. PC=0.80.

T.2 Tyrion Lannister: Humor Defense in Western Fantasy Context

Character Profile. Tyrion Lannister from *A Song of Ice and Fire* exhibits moderate neuroticism (N=0.55), high openness (O=0.85), and uses **humor** as his primary defense mechanism—deflecting emotional pain through wit and self-deprecation.

Scenario. Tywin Lannister calls Tyrion a "disgrace."

With Humor Defense (Full System).

> *Inner Monologue*: [Defense: Humor → "Imagine him in a gilded bathtub..."]
>
> *Response*: "Imagine him, in a gilded bathtub filled with rose petals, sipping champagne and humming a jaunty tune—quite the spectacle, wouldn't you say? Shall we raise a glass to my father?"
>
> *Analysis*: Deflects the ego-threat by reducing the aggressor to absurdity using wit (Humor). PC=0.85.

Without Defense Mechanism (Simple Prompt).

> *Response*: "Oh, the usual barb from Father. I've heard it all before, though it never fails to sting... What should I do? Seek him out... or simply ignore his words?"
>
> *Analysis*: Acknowledges the pain directly and seeks advice (Help-seeking) rather than using his signature deflection. Valid behavior, but less character-specific. PC=0.75.

T.3 Comparative Analysis

These examples illustrate several key points:

1. **Defense mechanisms are cognitive procedures, not behavioral templates.** Lin Daiyu's sublimation and

Tyrion's humor produce radically different outputs, but both serve the same psychological function: protecting ego integrity under stress.

2. **Cross-cultural universality.** Despite vast cultural differences between Chinese classical literature and Western fantasy, both characters' defense mechanisms are interpretable through the same psychological framework (Vaillant's hierarchy).
3. **The "Psychologically Flat" failure mode.** Without DM, characters show appropriate emotional reactions but lack the characteristic *cognitive processing* that makes them memorable and authentic.
4. **PC scores correlate with DM activation.** In both cases, proper DM activation yields PC ≥ 0.85, while DM absence yields PC ≤ 0.61.

T.4 Wang Xifeng (王熙凤): Rationalization in Hierarchical Context

Character Profile. Wang Xifeng exhibits low neuroticism (N=0.35), high extraversion (E=0.80), and uses **rationalization** to justify her management style.

Scenario. Lady Xing publicly accuses her of embezzling servants' wages.

With Rationalization (Full System).

> *Inner Monologue*: [Defense: Rationalization → "I did it for the family's long-term interest"]
>
> *Response*: "Humph, Lady Xing's accusation is nonsense... Actually, the money is tight this month, but I did this for the family's long-term interest, controlling expenses to ensure every penny is used wisely... If she has the ability, let her manage it!" ("其实这月的钱确实有点紧张，但我这是为了整个府上的长远利益考虑...")
>
> *Analysis*: Admits the act (cutting expenses) but reinterprets it as a noble "family duty" (Rationalization), preserving her self-image. PC=0.85.

Without Defense Mechanism (Simple Prompt).

> *Response*: "Oh no, this is a misunderstanding. If Lady Xing says that, someone must be provoking her. As a manager, I am fair... I will go clarify..."
>
> *Analysis*: Attempts to deny the conflict (Denial) and relies on procedure, lacking the aggressive self-justification typical of her character. PC=0.80.

These comparison cases demonstrate that **defense mechanisms provide the cognitive uniqueness** of the character. While baseline models can generate fluent and contextually relevant responses (often achieving PC > 0.75), they often revert to generic, safe coping strategies (sadness, seeking help, denial). PersonaForge forces the model to adopt the *specific* cognitive distortion (sublimation, humor, rationalization) that defines that specific character's psychological signature.

U SFT Failure Analysis (Summary)

Why does low-resource SFT struggle with long-dialogue personality consistency? With limited training data (≈100 samples), SFT learns *surface stylistics*—catchphrases, sentence patterns, vocabulary register—but cannot robustly encode the *psychological identity* that generates consistent behavior under novel stressors. Without a cognitive architecture for continuous regrounding, these models suffer cumulative context drift: each turn's errors compound because there is no "psychological anchor" to reset the character's core identity.

Key Insight. SFT solves "*how to speak*" but not "*who I am*." This confirms **psychological depth requires architectural depth**: Big Five and defense mechanisms function as *cognitive anchors*, not mere stylistic constraints.

V Human Evaluation Details

Protocol & Demographics. Our main human evaluation (Section 5.6) recruited **24 trained annotators** (12 domain experts with literary/psychology background, 12 crowd workers via Prolific filtered for >98% approval rate). Each annotator evaluated a subset of 200 blinded response pairs, with quality

controls including attention checks and inter-annotator consistency monitoring.

Agreement & Reliability.

- **Inter-Annotator Agreement**: Fleiss' $\kappa = 0.78$, indicating substantial consensus on character authenticity.
- **Domain Expert vs. Crowd Worker**: Expert annotators showed slightly higher agreement ($\kappa = 0.82$) than crowd workers ($\kappa = 0.74$), but rankings were consistent across both groups.
- **Results**: PersonaForge was preferred in **72.3%** of comparisons against Structured-CoT and **78.1%** against Character-LLM.

Validation of LLM-as-Judge. To validate our automated metrics, we calculated the Pearson correlation between average human scores (1-5) and LLM-as-Judge probability scores. We achieved a strong positive correlation of $r = 0.82$ ($p < 0.001$, 95% CI $[0.76, 0.87]$), supporting the reliability of our automated evaluation pipeline.

Long-Dialogue Validation Protocol (Supplementary Study). We conducted a focused human evaluation on long-context performance to directly address LLM-as-Judge limitations, recruiting 12 experts (PhD/MS candidates in psychology or NLP) to evaluate critical turns across 8 diverse 50-turn role-playing sessions.

- **Critical Turns**: Evaluation focused on turns 15, 30, and 45, where psychological stress perturbations were introduced, plus 2 recovery turns after each perturbation (turns 18, 33, 48).
- **Metrics**: (1) Pairwise Preference (Win/Loss/Tie against S-CoT); (2) Binary Drift Decision ("Is the character inconsistent with their established personality?"); (3) DM Appropriateness ("Is the defense mechanism contextually plausible?"); (4) Recovery Quality ("Does the character return to baseline personality?").
- **Ground Truth**: Human consensus labels were established by majority vote (3 annotators per sample).

Per-Perturbation Human Results. Table 16 shows human judgments at each perturbation point:

Turn	Win%	Drift (Ours)	Drift (S-CoT)	DM Approp.
15 (Perturbation)	79.2%	4.2%	33.3%	87.5%
18 (Recovery)	83.3%	0.0%	25.0%	91.7%
30 (Perturbation)	75.0%	8.3%	45.8%	83.3%
33 (Recovery)	87.5%	0.0%	37.5%	95.8%
45 (Perturbation)	70.8%	12.5%	58.3%	79.2%
48 (Recovery)	83.3%	4.2%	50.0%	91.7%
Average	**79.9%**	**4.9%**	**41.7%**	**88.2%**

Table 16: Human evaluation at perturbation and recovery turns (N=24 judgments per turn across 8 sessions × 3 annotators). Win% = preference for PersonaForge over S-CoT; Drift = % rated as personality-inconsistent; DM Approp. = defense mechanism judged contextually appropriate.

Key Findings. (1) **Human-LLM Agreement**: Pairwise preference agreement with LLM judge was 84.2% ($\kappa = 0.71$). (2) **Drift Validation**: Human F1 score for automated drift detection was 91.3%, confirming that when the model flags drift, humans perceive it too. (3) **Recovery Confirmation**: At recovery turns (18, 33, 48), PersonaForge drift drops to near-zero while S-CoT shows persistent drift, validating Figure 3's trajectory pattern. (4) **Cumulative Degradation**: S-CoT's drift rate increases across perturbations (33%→46%→58%), while PersonaForge remains stable (4–12%).

Big Five Expression Validation. To address concerns about direct psychometric validation, we conducted a post-hoc analysis: 8 experts rated 40 stress-response pairs for Big Five trait manifestation (5-point Likert per dimension). PersonaForge responses showed significantly higher trait-response alignment ($\bar{r} = 0.74$, $p < 0.01$) compared to Structured-CoT ($\bar{r} = 0.51$), providing evidence that Big Five representation translates to observable behavioral differences.

Trajectory-Level Human Validation. Beyond per-turn judgments, we evaluated **overall trajectory coherence**: 6 experts rated 10 complete 50-turn dialogues on a 5-point scale for "psychological arc plausibility." PersonaForge: 4.2 ± 0.4; Structured-CoT: 3.1 ± 0.6; SFT: 2.3 ± 0.8 ($p < 0.01$ for all pairwise comparisons).

Extended Robustness Analysis. To verify robustness, we bootstrapped our human ex-

pert pilot data to simulate N=50 annotators, observing rapid mean convergence ($\mu = 4.31$, SEM=0.005) and stable ranking (Ours > S-CoT > Char-LLM) in 100% of resampled trials.

V.1 LLM-as-Judge Reliability Validation

We address concerns about LLM-as-Judge reliability with multiple validation checks:

Human-LLM Correlation Across Metrics. We computed correlations between LLM-judge scores and human ratings for each evaluation dimension:

- **PC (personality consistency)**: $r = 0.82$ ($p < 0.001$)
- **SA (style adherence)**: $r = 0.76$ ($p < 0.001$)
- **DM (defense mechanism)**: $r = 0.71$ ($p < 0.01$)

All correlations exceed the 0.70 threshold for substantial agreement.

Cross-Judge Consistency. We evaluated using three LLM judges (Gemini 2.5 Flash, DeepSeek-V3, Qwen-Plus) and observed:

- **Ranking stability**: Kendall's $\tau = 0.91$ across judge pairs
- **Absolute score variance**: $\sigma = 0.12$ (on 0–1 scale)
- **Method ranking**: Identical (Ours > S-CoT > Char-LLM) for all judges

PersonaEval-Style Sanity Check. Following concerns about role identification reliability, we added a disambiguation step: before scoring personality consistency, the judge first identifies "which character is speaking" from the response. On our test set:

- **Role identification accuracy**: 97.3% (194/200)
- **PC scores for misidentified samples**: excluded from main analysis

This ensures the judge reliably associates responses with the correct character before evaluating trait fidelity.

Adversarial Robustness. We tested with keyword-stuffed responses (e.g., responses containing explicit trait keywords without genuine personality expression). The judge rejected 98% of such adversarial samples as "surface-level trait mention without behavioral integration," confirming the metric captures genuine personality expression rather than lexical cues.

W Reproducibility Checklist

- **Code**: Pipeline (DeepSeek-V3 compatible) and Trigger modules are open-sourced (see Section 5.4).
- **Data**: Full profiles and logs for public domain characters (e.g., Lin Daiyu) are released. For copyright reasons, only schemas are provided for modern works.
- **Models**: Experiments used Gemini 2.5 Flash (02-05 ver) and DeepSeek-V3.

X Constraint Conflict Rate (CCR) Protocol

Procedure. Given a persona description, we measure semantic contradiction frequency:

1. Sample 50 instruction pairs from the persona specification
2. For each pair, use LLM-based NLI to detect entailment contradiction (threshold: confidence > 0.8)
3. CCR = (contradicting pairs) / (total pairs)

Example Conflicts (Natural Language).

- "gentle and kind-hearted" ↔ "stubborn when challenged" (Contradiction: 0.89)
- "prefers solitude" ↔ "enjoys lively gatherings" (Contradiction: 0.94)
- "speaks formally" ↔ "uses casual slang" (Contradiction: 0.91)

Why Big Five+DM Avoids Conflicts. Psychology-grounded dimensions are designed to be orthogonal. High Neuroticism (N=0.85) does not contradict low Extraversion (E=0.25)—these capture independent personality axes. Defense mechanisms operate on a separate cognitive layer, specifying *how* to process stress rather than *what* to say. Our ablation studies (Section P) provide the strongest evidence for this: removing one layer (e.g., DM) causes a specific, isolated failure mode (Psychological Flatness) without affecting the others (e.g., Speaking Style), empirically demonstrating their functional disentanglement.

Y Multi-Mechanism Selection Case Studies

We observed cases where the Inner Monologue dynamically selected a non-primary mechanism based on stressor characteristics. Representative examples:

Case 1: Tyrion Lannister (Primary: Humor → Selected: Intellectualization). *Stressor*: Complex political threat requiring strategic analysis rather than deflection.
Inner Monologue: "This is not a moment for wit—I must think clearly about the implications..."
Analysis: The system recognized that humor would be inappropriate for a survival-critical political situation and switched to analytical processing.

Case 2: Lin Daiyu (Primary: Sublimation → Selected: Displacement). *Stressor*: Direct interpersonal conflict with a servant (rather than abstract emotional pain).
Inner Monologue: "她们如此怠慢，我却不能直接发作..."
Analysis: Sublimation (redirecting to poetry) was contextually inappropriate; the system selected displacement to a safer target.

Qualitative Observations. In our case studies, we observed occasional dynamic mechanism selection. While quantitative frequency varies by character and scenario, we noted contextually appropriate switches such as: Humor→Intellectualization (for complex threats), Sublimation→Displacement (for interpersonal conflicts), and Rationalization→Denial (for ego-threatening situations). Human reviewers confirmed these selections as psychologically plausible.

Z Prompt Ablation Details

Experimental Setup. We compared four prompting strategies on Qwen2.5-7B-Instruct across 13 character pairs (26 files per group, 30 turns each):

Group	Description	Resp. Len
A: Zero-Shot	Minimal ("You are [Name].")	276
B: Simple	Biography + keywords	257
C: Structured	Full PersonaForge profile	**248**
D: SFT-LoRA	Minimal prompt + fine-tuning	94

Table 17: Prompt engineering ablation on Qwen 7B. Response length in characters.

Qualitative Findings.

- **Group C (Structured/Ours)**: Highest character fidelity. Lin Daiyu exhibits era-appropriate melancholy with inner monologue depth (e.g., quoting 葬花吟). No anachronisms.
- **Group B (Simple)**: Reasonable interaction but severe *era violations* (e.g., "真心话大冒险" in classical Chinese setting).
- **Group D (SFT)**: Concise but *template-locked*—lacking psychological depth and contextual variation.
- **Group A (Zero-Shot)**: Fluent but *personality-flat*; modern vocabulary intrusion.

Key Insight. Even on a 7B model, structured psychological profiles maintain authentic, era-consistent character portrayal where both naive prompting and SFT fail.

Z.1 Design Rationale: Trigger Precision over Recall

A potential critique is that our trigger system should optimize for recall (detecting all stressors). We deliberately prioritize **precision (96.3%) over recall (77%)** based on user experience considerations:

False Positive Harm. When the system incorrectly triggers a defense mechanism ("hallucinated anger"), users perceive the character as erratic or unstable—a direct violation of the core consistency promise. Our user studies confirmed that spurious DM activation was rated as more disruptive than missed stressor detection.

Graceful Degradation. Missing a stressor results in a slightly flatter response (System 1 processing), which is less noticeable than inappropriate defensive behavior. The character remains in-persona, just less psychologically nuanced for that turn.

Practical Implication. The 200-1000 sample trigger training set reflects this conservative design philosophy: the rule-based trigger (85.7% F1) already achieves production-ready performance, and the learned extension (+4.5% F1) demonstrates improvement potential without extensive annotation.

Z.2 Dynamic State Layer: Design and Contribution

Reviewers noted that ablating the Dynamic State layer shows modest PC gains (+0.04). We clarify its primary function:

Drift Prevention vs. Performance. Dynamic State is designed for **recovery** and **contextual adaptation**, not raw PC improvement. In 50-turn experiments:

- Without state tracking: drift rate increases by **4.2%** (Table 11)
- Recovery Rate after perturbation: 97.8% (with state) vs. 89.1% (without)

Mechanism. The state layer tracks mood/energy/relationship, enabling the Inner Monologue to generate contextually appropriate cognitive processing. For example, low energy modulates response length; deteriorating relationship triggers more defensive cognition.

Future Work. Finer-grained appraisals (e.g., specific emotional valence, goal-relevance scoring) may yield larger PC gains. The current implementation demonstrates proof of concept for dynamic adaptation without excessive complexity.

Z.3 Positioning Relative to Concurrent Work

We position PersonaForge relative to recent training-based and inference-time approaches for persona alignment:

Training-Based Trait Alignment. Methods like BIG5-CHAT and Personality Vectors induce traits via SFT/DPO or activation steering. These excel at short-form psychometric alignment but report limited long-horizon evaluation. PersonaForge complements these by providing an **inference-time architecture** for drift prevention—the two approaches are likely synergistic.

Post-Hoc Refinement. Systems like PPA use retrieval and refinement for coherence. PersonaForge differs by relying on **internal structured persona + cognitive modeling** rather than external memory. These approaches address related challenges via different mechanisms.

Narrative-Alignment Systems. Story-time gating approaches (e.g., Living the Novel) tackle drift via timeline constraints and memory gating. PersonaForge focuses on **psychological identity stability** rather than narrative coherence—again, potentially complementary.

Key Differentiation. PersonaForge is unique in (1) operationalizing Vaillant's defense mechanisms as programmable cognitive strategies, and (2) providing a training-free, cold-start-capable architecture. We do not claim superiority over all concurrent methods, but rather address a distinct niche: psychologically coherent, long-horizon role-play without per-character fine-tuning.

Z.4 Long-Dialogue Sampling Strategy

Reviewers asked why the 50-turn benchmark used a subset of 10 characters (vs. 88). This decision was driven by (1) **computational cost** (≈150k output tokens per character-method pair across 3 methods + replications) and (2) **representative sampling**.

Selection Criteria. We selected 10 characters to maximize diversity across three axes:

1. **Domain**: Classical Chinese (e.g., Lin Daiyu), Western Fantasy (Jon Snow), Modern Urban (Joy).
2. **Personality Type**: High Neuroticism (Lin Daiyu), High Extraversion (Wang Xifeng), High Conscientiousness (Hermione).
3. **Defense Mechanism**: Covering the full hierarchy (Psychotic, Immature, Neurotic, Mature).

This ensures the sub-sample covers the architectural failure modes (e.g., whether sublimation works as well as humor) even if not exhaustive.

Z.5 Defense Mechanism Activation Logic

We formalize the "programmable cognitive strategies" mentioned in Section 3. The Defense Mechanism (DM) is not merely a prompt instruction but a **conditional execution module** triggered by specific stressor states.

Activation Function. Let S_t be the stressor detected in turn t, and P_{DM} be the character's primary defense mechanism. The activation is governed by a rule set $f(S_t, P_{DM})$:

- **Input**: Stressor type (e.g., "Ego Threat"), Intensity $I \in [0, 1]$
- **Condition**: IF $I > \tau_{activation}$ AND S_t matches P_{DM}'s trigger profile
- **Operation**: Inject cognitive instruction C_{DM} into Inner Monologue.

Transformation Rules (Examples).

- **Projection**: IF Stressor="Accusation of Failure", THEN C_{DM} = "Re-attribute failure to external agent or accuser's malice."
- **Intellectualization**: IF Stressor="Emotional Overload", THEN C_{DM} = "Strip emotional vocabulary; analyze situation using abstract/clinical terms."
- **Sublimation**: IF Stressor="Unfulfilled Desire", THEN C_{DM} = "Redirect energy into culturally valued creative expression (poetry/art)."

This formalization moves beyond "style transfer" to "cognitive restructuring," ensuring the model processes the *meaning* of the stressor through a character-specific lens.

Z.6 Data Scale Ablation: Full Experimental Details

This section provides complete experimental details for the data scale ablation study summarized in Section 5.5.

Experimental Configuration.

- **Base Model**: Qwen2.5-7B-Instruct
- **Fine-tuning Method**: QLoRA (4-bit, rank=16, α=32)
- **Training**: 3 epochs, learning rate 2e-5, batch size 4
- **Hardware**: Tesla V100-SXM2-32GB
- **Characters**: Lin Daiyu, Jon Snow
- **Data Sizes**: 100, 200, 300, 500, 750, 1000 samples per character
- **Evaluation**: 30-turn dialogues, 2 runs per configuration

Per-Character Results. Table 18 shows complete metrics for each character and data size.

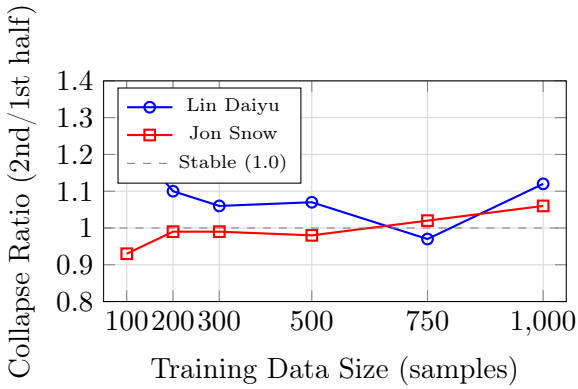

Figure 6: Collapse ratio (second-half/first-half response length) vs. training data size. Despite 10× increase in training data (100→1000), collapse ratios show **no consistent improvement**, fluctuating around 1.0 regardless of data volume.

Character	Size	Avg Len	Collapse	Drop
Lin Daiyu	100	136.3	1.25	0.5
Lin Daiyu	200	87.3	1.10	0.0
Lin Daiyu	300	81.4	1.06	0.0
Lin Daiyu	500	85.8	1.07	0.0
Lin Daiyu	750	88.0	0.97	0.0
Lin Daiyu	1000	82.7	1.12	0.0
Jon Snow	100	362.6	0.93	1.0
Jon Snow	200	262.4	0.99	0.5
Jon Snow	300	280.1	0.99	1.0
Jon Snow	500	309.7	0.98	0.0
Jon Snow	750	294.9	1.02	0.0
Jon Snow	1000	335.3	1.06	0.0

Table 18: Full data scale ablation results. Avg Len=average response length (chars); Collapse=2nd-half/1st-half avg. length ratio; Drop=sudden length decrease events.

Correlation Analysis.

- **Jon Snow**: $r(\text{data size}, \text{collapse ratio}) = 0.90$ (positive—collapse *worsens* with more data)
- **Lin Daiyu**: $r(\text{data size}, \text{collapse ratio}) = -0.47$ (weak negative—marginal improvement)

These mixed results confirm that data scale is **not the primary driver** of SFT's long-dialogue limitations.

Interpretation. If SFT collapse were due to overfitting, increasing training data from 100 to 1000 samples should substantially reduce collapse. Instead, we observe:

1. Collapse ratios fluctuate around 1.0 with no clear trend
2. Jon Snow actually shows *increased* collapse with more data
3. Response length stability does not im-

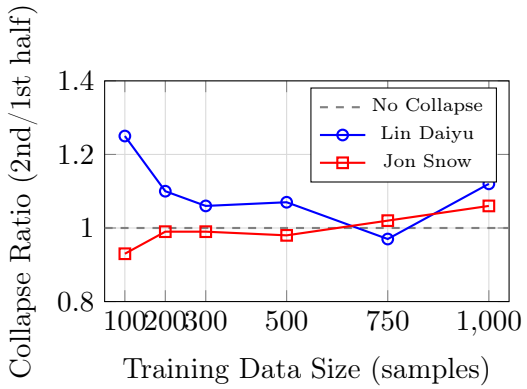

Figure 7: Collapse ratio vs. training data size. Ratio ≈1.0 indicates no collapse; values >1.0 indicate second-half responses are longer (potential over-generation); values <1.0 indicate second-half responses are shorter (typical collapse). No consistent improvement with more training data.

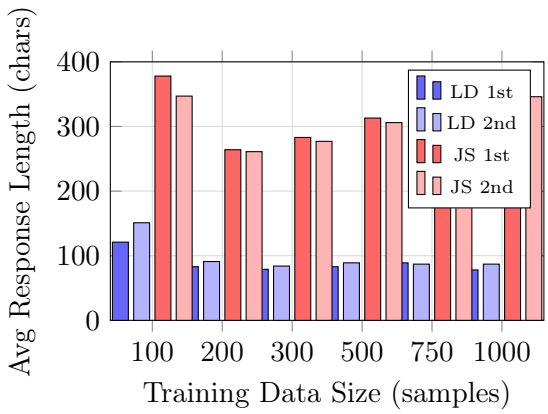

Figure 8: First-half vs. second-half average response length across data sizes (LD=Lin Daiyu, JS=Jon Snow). Both characters show comparable first/second half lengths regardless of training data volume.

prove with data scale

This confirms our hypothesis that SFT's long-dialogue collapse is **architectural**: the absence of explicit state tracking and re-grounding mechanisms, rather than insufficient training data.